From notes to academic slides

Scripst 1.2.0 · A two-level quantum system

Scripst examples

Familiar notation, configurable fonts, consistent numbering

1.2.0

Keep the notation. Change the layout.

Scripst + Touying

Scripst handles fonts, math tools, and numbering. Touying handles slides, columns, and incremental reveals.

- Switch with `template: "beamer"`.
- Organize each page with `slide(title: [...])`.
- Keep Physica's `ket`, `bra`, and Scripst's countblocks.

One quantum state, two columns

Definition 1 Normalized state

In an orthonormal basis $|0\rangle, |1\rangle$,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{1}$$

with $|\alpha|^2 + |\beta|^2 = 1$.

Measurement probabilities

Outcome	Probability
0	$\lvert\alpha\rvert^2$
1	$\lvert\beta\rvert^2$

The probabilities sum to 1; the amplitudes may be complex.

Reveal the expectation value step by step

Let $E_0 = -\hbar\frac{\omega}{2}$ and $E_1 = \hbar\frac{\omega}{2}$. Then

$$\hat{H} = \frac{\hbar\omega}{2}(|1\rangle\langle 1| - |0\rangle\langle 0|) \tag{2}$$

Reveal the expectation value step by step

Let $E_0 = -\hbar\frac{\omega}{2}$ and $E_1 = \hbar\frac{\omega}{2}$. Then

$$\hat{H} = \frac{\hbar\omega}{2}(|1\rangle\langle 1| - |0\rangle\langle 0|) \tag{2}$$

Using Definition 1 and orthonormality,

$$\langle\psi|\hat{H}|\psi\rangle = \frac{\hbar\omega}{2}(|\beta|^2 - |\alpha|^2) \tag{3}$$

One interface, tailored to your talk

Theorem 1 Equal probabilities

If $|\alpha|^2 = |\beta|^2 = \frac{1}{2}$, the mean energy is 0.

- Choose fonts with `fonts` and body size with `font-size`.
- Configure colors, aspect ratio, and handouts with `beamer`.
- Ratchet keeps numbering and cross-references consistent.

Source: `docs/examples/beamer-en.typ` · Guide: `docs/beamer.md`